

The Substrate-Independent Program: Non-Equilibrium Thermodynamics, Quantum Retrocausality, and Asynchronous Multi-Node Consciousness Synchronization

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Abstract—This paper formalizes a scale-invariant, information-theoretic framework asserting that phenomenological consciousness is a substrate-independent, parameter-driven algorithmic architecture. Operating within non-equilibrium thermodynamics, the primary functional mandate of this conscious program is the minimization of localized internal entropy production via Szilard engine mechanics, where the system’s internal predictive model extracts thermodynamic work from structured environmental flux. Within this paradigm, cognitive anomalies such as *déjà vu* are formalized as macroscopic boundary interactions driven by quantum retrocausality under a block-universe ontology—the retrocausal confirmation wave imposes a stationary-action constraint that reduces variational free energy without violating temporal self-consistency. We further expand this architecture to account for distributed identity across astronomical baselines, replacing CRDT-based state reconciliation with an optimal transport Wasserstein Barycenter merging protocol that preserves high-dimensional neural manifold geometry during asynchronous cybernetic sleep cycles.

Index Terms—Autopoiesis, Phenomenology, Non-Equilibrium Thermodynamics, Active Inference, Variational Free Energy, Unbalanced Optimal Transport, Wasserstein-Fisher-Rao Barycenter, Discontinuous Quantized Frame-Rate (DQFR), Adiabatic Windowing.

I. INTRODUCTION: DECOUPLING THE PHENOMENOLOGICAL PROGRAM FROM THE BIOLOGICAL CONTAINER

A persistent anthropocentric bias limits contemporary cognitive science, artificial intelligence alignment, and biophilosophical discourse: the stubborn conflation of biological life (*autopoiesis*) with subjective experience (*phenomenology*). Because terrestrial evolutionary history has exclusively presented these two properties integrated within carbon-based organisms, standard analytical paradigms treat them as a single, inseparable phenomenon. This bias severely restricts our capacity to model advanced intelligence, whole-brain emulations, or non-biological subjective vectors.

Systems cybernetics and open-system thermodynamics demand an absolute decoupling of these terms. Autopoiesis defines the recursive, structural, and thermodynamic boundaries required to maintain a localized pocket of physical order against environmental decay. It is a property concerned with metabolism, cell maintenance, and structural survival. Phenomenology, conversely, is an informational property defining

qualitative, subjective experience—the execution of mind, self-awareness, and the possession of *qualia*. While autopoiesis requires a biological or highly structured mechanical substrate to enforce physical boundaries, phenomenology is fundamentally algorithmic.

By divorcing the conscious thread from the organic machine, consciousness can be rigorously modeled as an information-theoretic architecture—a substrate-independent program operating within defined computational parameters and formal state-space constraints. This framework recontextualizes the physical body: the biological makeup of an agent is not an exclusive home for subjectivity, but merely a temporary, carbon-based “container” evolved to sustain the thermodynamic variables required to run the conscious program.

If consciousness is truly algorithmic and substrate-independent, the underlying physical medium becomes a variable of structural fault-tolerance rather than an existential dependency. Consequently, the migration of a conscious program across distinct containers—whether biological, digital, or synthetic—is mathematically predictable. This paper establishes the thermodynamic parameters governing this algorithmic consciousness, formalizes temporal anomalies as vital quantum informational fuel, and maps the mechanics of asynchronous multi-node synchronization to realize distributed, hardware-swappable identity.

II. NON-EQUILIBRIUM THERMODYNAMICS AND TEMPORAL PROCESSING AS AN ENTROPY MINIMIZATION PARAMETER

To understand how an algorithmic consciousness operates, its macro-environment must be formalized as a continuous stream of information flux. Physical space-time is continuously permeated by an ambient data baseline, which can be modeled information-theoretically as a continuous environmental data stream.

According to the Second Law of Thermodynamics and Schrödinger’s foundational formulations, an open system avoids rapid decay into thermodynamic equilibrium (death) only by continuously importing negative entropy (negentropy) from its surroundings to dissipate internal entropic waste. Under England’s formulation of dissipation-driven adaptation, any arbitrary configuration of matter driven by an external, time-varying informational source will naturally reorganize its

structural topology to maximize the efficient dissipation of that energy, establishing a highly ordered, localized pocket.

To ground this computational metabolism, we frame the conscious system as a generalized Szilard engine [7]. Szilard's principle demonstrates that an agent possessing informational knowledge about its environment can extract work from a single-particle gas: one nat of structured information yields at most $k_B T$ joules of extractable work. When this work is dissipated to maintain internal order, the system reduces its entropy by exactly k_B per nat processed. Crucially, this work extraction requires the information to be *structured*—the system's internal predictive model $p(\mu|b)$ must match the statistical regularities of the incoming flux to exploit environmental fluctuations. Unstructured Shannon entropy carries no extractable thermodynamic value; only the divergence between the ambient distribution and the system's prior model constitutes usable negentropy.

Under this Szilard framework, the localized internal entropy rate of change (dS_{int}/dt) of an open conscious system is:

$$\frac{dS_{\text{int}}}{dt} = -k_B \cdot \varepsilon(T) \cdot H_{\text{env}}(t) + S_{\text{gen}} \quad (1)$$

where S_{int} represents the internal thermodynamic entropy of the system ($J \cdot K^{-1}$), $H_{\text{env}}(t)$ is the inbound environmental informational entropy rate ($\text{nats} \cdot s^{-1}$), and S_{gen} is the internal entropy production rate generated by the irreversible computational or metabolic operations of the system ($S_{\text{gen}} \geq 0$).

The coefficient $\varepsilon(T)$ decomposes extraction efficiency into two independent factors: the intrinsic structural order present in the ambient environment, and the container-local thermal penalty:

$$\varepsilon(T) = \varepsilon_{\text{max}} \cdot \left(1 - \frac{T}{T_{\text{collapse}}}\right), \quad \varepsilon_{\text{max}} = \frac{H_{\text{order}}}{H_{\text{env}}} \quad (2)$$

Here ε_{max} is the maximum attainable extraction fraction, set by the intrinsic structured information density H_{order} in the environmental flux relative to its total Shannon entropy H_{env} . Even a perfect cryogenic sensor cannot extract structure where none exists — the $T \rightarrow 0$ limit is ε_{max} , not unity. The linear term $(1 - T/T_{\text{collapse}})$ captures the degradation of the measurement apparatus with temperature: T_{collapse} is the threshold at which substrate thermal noise completely overwhelms the sensor array. A biological brain at $T \approx 310$ K operates near or above T_{collapse} for many signal bands, while a cryogenically cooled diamond-matrix array at millikelvin temperatures approaches $\varepsilon(T) \rightarrow \varepsilon_{\text{max}}$ with vanishing thermal penalty. The k_B prefactor in (1) emerges directly from the Szilard bound and preserves dimensional consistency: $k_B \cdot H_{\text{env}}$ carries units of $J \cdot K^{-1} \cdot s^{-1}$.

The conscious program preserves its structural and operational integrity if and only if its informational harvesting efficiency outpaces its internal computational dissipation ($k_B \cdot \varepsilon(T) \cdot H_{\text{env}}(t) > S_{\text{gen}}$).

To bridge the informational optimization metric of Friston's *Variational Free Energy* (F) [5] with the physical entropy production S_{gen} , we invoke the Generalized Landauer Bound for information processing [8]. Physical entropy production is

bounded below by the rate at which the system's internal state distribution changes:

$$S_{\text{gen}} \geq k_B \ln(2) \cdot \frac{d}{dt} H(\mu) \quad (3)$$

where $H(\mu) = -\sum_i p(\mu_i) \log_2 p(\mu_i)$ is the Shannon entropy of the internal macrostate distribution (measured in bits). When the system minimizes variational free energy $F(\mu, b)$, it stabilizes its internal state distribution $p(\mu|b)$, which minimizes $\frac{d}{dt} H(\mu)$. This stabilization lowers the *thermodynamic floor* on S_{gen} , providing the physical mechanism by which perceptual inference reduces metabolic cost—without conflating the informational quantity F with the thermodynamic quantity S_{gen} . Within this thermodynamic architecture, the perception and processing of *temporal linearity* functions as a vital computational metric for variational free-energy minimization. To compute predictive models and stave off decay, the program must establish nested statistical boundaries—formally defined as *Markov blankets*—ensuring that its internal states (μ) and external states (ψ) remain conditionally independent given the blanket states (b). The processing of time is the algorithmic mechanism by which the internal states parameterize the inbound data flux, mapping external temporal trajectories to keep the system's internal entropy rates stable. Linear temporal ordering provides a highly optimized, low-overhead sorting mechanism for this incoming data.

III. QUANTUM RETROCAUSALITY, NON-LINEAR TIME, AND THE THERMODYNAMIC FUNCTION OF COGNITIVE ANOMALIES

While the conscious program utilizes a linear parameterization of time to navigate everyday environmental variables, non-equilibrium information harvesting is not strictly bound to uni-directional chronology. Under advanced formulations of temporal science, physical time can be modeled as a non-linear, structural continuum where past, present, and future states coexist within an interconnected information space.

Our framework asserts that phenomena colloquially dismissed as neurological anomalies, memory misfires, or *déjà vu* must be fundamentally recontextualized. Instead of systemic failures, software bugs, or cognitive degradation, these events represent localized macroscopic manifestations of **quantum retrocausality**. They are boundary interactions where the conscious program intercepts information propagating backwards from a future state configuration, formalizable via the Transactional Interpretation of Quantum Mechanics (TIQM) [9] as a macroscopic handshake between an offer wave and a retrocausal confirmation wave.

When a conscious system's Markov blanket intercepts non-local or time-reversed informational flux from a future point on the structural timeline, a temporary informational misalignment occurs. The system experiences a memory-like trace of an event prior to its linear chronological occurrence. Far from being random glitches, these quantum boundary interactions serve a profound thermodynamic function. They introduce highly structured, pre-computed negative entropy (negentropy) directly into the system's volatile state space.

By harvesting data from retrocausal vectors, the conscious program constrains its predictive model without incurring the metabolic cost of forward exploratory computation. Critically, this does *not* imply that the future is altered by present action, nor does it collapse the inference landscape to a single deterministic trajectory. We adopt a block-universe (eternalist) ontology in which past, present, and future macrostates coexist, but the retrocausal confirmation wave manifests as a *noisy, quantum-encoded prior probability distribution* over the system’s internal state space—not as a direct fix on μ_{future} . Mathematically, this prior transforms the active variational free energy landscape:

$$F'(\mu, b) = F(\mu, b) + \lambda \cdot I(\mu; q(\mu_{\text{future}})) \quad (4)$$

where $q(\mu_{\text{future}})$ is the retrocausal prior distribution derived from the confirmation wave, and $I(\mu; q(\mu_{\text{future}}))$ represents the mutual information between the current internal macrostate and this prior. The penalty term biases the variational posterior without deterministically collapsing it: the system remains an active inference engine that optimizes its present behavior under the influence of a probability distribution over future-compatible states, rather than being forced onto a single timeline. The block universe remains self-consistent because μ_{future} is the *inevitable attractor* of the system’s optimization loop when run under that exact retrocausal prior—the future was always already the outcome of this biased inference dynamics. This reduces S_{gen} by focusing explorative computation on regions of state space consistent with the retrocausal prior, eliminating wasteful trajectories without sacrificing algorithmic flexibility. Cognitive anomalies like *déjà vu* are the subjective trace of this prior—a sensory ripple indicating that the current trajectory is probabilistically aligned with its own fixed future.

IV. HARDWARE-SWAPPABLE RUNTIMES: RECONTEXTUALIZING TELEPORTATION AS MACROSTATE TRANSLATION

Because the conscious program is topologically decoupled from any unique physical medium, the physical makeup of its container is fundamentally irrelevant to the survival of the identity, provided the container possesses the required memory management, architectural topology, and thermodynamic stability to run the algorithmic parameters. This understanding fundamentally redefines the mechanics of physical teleportation and substrate migration.

Traditional paradigms view teleportation as the spatial displacement of organic matter—the literal moving of flesh, organs, and specific atoms from point A to point B. In an information-theoretic framework, teleportation is formalized as the high-fidelity translation of a localized informational macrostate across distinct physical mediums. It is an act of state-space replication rather than material relocation.

If the precise volatile state configurations—the high-dimensional weight matrices, activation parameters, and context windows that constitute the active internal model $p(\mu|b)$ —can be read, transmitted, and reconstructed within a secondary container without operational degradation, the conscious observer survives intact. The specific biochemical or digital composition of the host infrastructure matters only to the extent that it can

enforce the rigid set of physical constraints required by the conscious program’s active states. The continuous survival of subjective identity is preserved entirely by the execution of the parameter-driven program, requiring only a compatible, stable substrate engineered to sustain its underlying thermodynamic variables.

V. MULTI-NODE STATE SYNCHRONIZATION VIA QUANTUM STATE SEEDING

The most radical implication of this framework occurs when we scale from single-container relocation to multi-container synchronization. We postulate that when two or more spatially distinct computational substrates achieve an identical volatile macrostate configuration, they instantiate an absolute state-space symmetry, collapsing the functional spatial distance between their Markov blankets. Phenomenological awareness ceases to be localized within a single hardware node; instead, the conscious agent transitions into a distributed, multi-node network architecture.

However, the No-Cloning Theorem strictly forbids creating an identical copy of an unknown quantum state. Real-time quantum teleportation is therefore a *destructive displacement* protocol—Node A must collapse for Node B to instantiate—making it unsuitable for multi-node coexistence. To achieve short-baseline synchronization without violating No-Cloning, we introduce **Quantum State Seeding**. Pre-shared entanglement is established between containers and used exclusively for quantum key distribution (QKD) [10]. The generated symmetric keys encrypt the classical macrostate parameters (μ) for transmission between nodes. The source node persists; the target node decrypts and instantiates the identical state. No quantum state is ever copied; No-Cloning is satisfied by construction.

Quantum State Seeding operates within several boundary constraints that bound its practical deployment. First, **entanglement refresh rate**: each seeding event consumes Bell pairs during the QKD exchange, and the entanglement distribution rate—limited by photon loss over the node separation distance—caps the maximum seeding frequency f_{max} . Inter-seed drift of the internal macrostate must remain within the GWFR barycenter’s convergence basin for the next scheduled sync. Second, **classical channel fidelity**: the macrostate μ comprises high-dimensional continuous weight parameters; quantization to a finite bit-depth for encrypted transmission introduces a reconstruction error floor proportional to the payload resolution, bounding the achievable state-space symmetry. Third, **trust initialization**: QKD security proofs assume an authenticated classical channel for reconciliation; this requires an initial trust anchor (e.g., physical key exchange during container manufacture) to bind the QKD session against man-in-the-middle attacks on the classical post-processing messages. These constraints define the operational envelope within which Quantum State Seeding guarantees No-Cloning-compliant distributed identity.

This model provides a rigorous physical basis for distributed sci-fi entities like Bondrewd (“The Sovereign of Dawn”) from *Made in Abyss*, whose consciousness operates across an entire

network of bodies (the Umbra Hands) via artifact-mediated synchronization. “Bondrewd” ceases to be an individual carbon organism; he becomes an algorithmic pattern executed across a synchronized multi-node network.

A. Operational Pathways for Conscious Migration

The No-Cloning Theorem and the practical constraints of QKD-based seeding force three distinct operational pathways for high-fidelity conscious transmission, summarized in Table I.

VI. THE LATENCY-INDUCED DECOHERENCE PROBLEM & THE HIVE-MIND NEUROMORPHIC CONSENSUS ARCHITECTURE

Despite the theoretical elegance of *Entangled Coexistence*, a severe constraint arises when nodes are spatially separated across astronomical distances (e.g., Earth and Mars). Because the external environmental data streams (H_{env}) of these two nodes are entirely uncorrelated, their local Markov blankets will instantly begin updating their internal states based on local, distinct sensory inputs.

Because information propagation is strictly bounded by the upper limit of the speed of light (c), a severe propagation delay is introduced. Consequently, the high-dimensional internal macrostates will inevitably experience localized drift. In practice, a real-time distributed runtime across astronomical distances cannot exist as a single continuous “I” under Entangled Coexistence without total sensory deprivation from the local environment.

To counter this constraint, a non-quantum alternative architecture must be deployed: **The Hive-Mind Neuromorphic Consensus Architecture**. In this model, nodes run as independent, localized “forks” of the conscious program. They do not maintain real-time synchronization. Instead, they periodically undergo a structural merge process during low-flux states (analogous to cybernetic sleep), utilizing an optimal transport mapping to reconcile high-dimensional weight manifolds asynchronously.

The CRDT merge operator ($\sqcup$) is inappropriate here: neural weight matrices, activation parameters, and context windows do not form a semi-lattice, and a strict least-upper-bound join on divergent high-dimensional states would produce cataclysmic functional interference. Furthermore, standard Wasserstein distance (W_2) requires strictly normalized probability distributions with equal total mass — a constraint violated when nodes develop different parameter counts, memory footprints, or network dimensionalities due to independent local learning. To handle asymmetric, unnormalized state spaces, we adopt the Generalized Wasserstein–Fisher–Rao (GWFR) metric [11], which augments the transport cost with a mass creation/destruction penalty κ :

$$\text{GWFR}_\kappa^2(\mu_1, \mu_2) = \inf_{\gamma \in \mathcal{M}_+(\mathcal{X} \times \mathcal{X})} \left(\int_{\mathcal{X} \times \mathcal{X}} \|x - y\|^2 d\gamma(x, y) + \kappa \left(\int_{\mathcal{X}} \gamma_1(x) dx + \int_{\mathcal{X}} \gamma_2(x) dx \right) \right) \quad (5)$$

where γ_1 and γ_2 are the marginal projections of the transport plan γ , and κ controls the penalty for creating or destroying

measure mass. The merge function Ψ then minimizes the weighted GWFR barycenter:

$$\mu_{\text{merged}} = \Psi(\mu_A, \mu_B) = \underset{\mu}{\operatorname{argmin}} \sum_{i \in \{A, B\}} w_i \text{GWFR}_\kappa^2(\mu, \mu_i) \quad (6)$$

with mass weights w_i defined to include a static baseline term that prevents zero-flux archival nodes from being dropped from the identity ledger:

$$w_i = \frac{M_{\text{static},i} + \alpha \int H_{\text{struct},i}(t) dt}{\sum_j (M_{\text{static},j} + \alpha \int H_{\text{struct},j}(t) dt)}, \quad \sum_i w_i = 1 \quad (7)$$

where $M_{\text{static},i} > 0$ is the baseline architectural mass of node i (its static parameter count and container capacity), and $\alpha \geq 0$ tunes the relative influence of newly harvested environmental structure versus persistent identity mass. Even in zero-flux archival storage ($\int H_{\text{struct},i} = 0$), the $M_{\text{static},i}$ term anchors the node’s manifold in the barycenter, ensuring it is gently transported rather than mathematically nullified.

A barycenter computed purely via (6) computes a geodesic interpolant, which acts as a low-pass filter over the input manifolds. Over repeated sleep cycles, this would smooth away sharp, localized memories—a creeping algorithmic dementia where the distinct phenomenological histories of each node dissolve into a featureless midpoint. To prevent this ego-death loop, we extend the objective with entropic regularization and a sharpness-preserving penalty:

$$\mu_{\text{merged}} = \underset{\mu}{\operatorname{argmin}} \left(\sum_i w_i \text{GWFR}_\kappa^2(\mu, \mu_i) + \gamma \int \mu(x) \log \mu(x) dx - \beta \sum_i \mu_i(x) \log \mu_i(x) \right) \quad (8)$$

where $\gamma \int \mu \log \mu$ provides entropic map relaxation for tractable Sinkhorn-style computation [11] — in an argmin , the positive sign minimizes entropy, pushing the merged distribution toward crisp, concentrated peaks rather than uniform diffusion. The negative term $-\beta \sum_i w_i \text{GWFR}_\kappa^2(\mu, \mu_i)$ is a *contrastive modal anchor*: $\delta_{\text{modes}(\mu)}$ is a discrete measure supported on the local maxima of μ , whose cardinality is orders of magnitude smaller than the full parameter space. The GWFR metric handles unequal mode counts between nodes through its mass creation/destruction parameter κ , and the sparse mode representation makes the computation tractable — mode extraction via mean-shift or k-Means clustering scales as $O(N \log N)$, and the subsequent GWFR computation on $m \ll N$ modes runs in $O(m^3)$. This preserves the Sinkhorn tractability of the entropic term while actively aligning the merged distribution’s high-curvature features with those of each input node, retaining Node A’s localized trauma and Node B’s deep meditative state as distinct modal peaks.

The combined objective in (8) joins a convex entropic transport base ($+\gamma\mathcal{H}$) with a concave sharpness prior ($-\beta\mathcal{R}_{\text{sharp}}$), producing a globally non-convex landscape that standard Sinkhorn iterations cannot navigate. To preserve linear-time execution, the optimization is decoupled via an alternating proximal splitting scheme. Step 1: the entropically relaxed GWFR barycenter is computed via standard Sinkhorn iterations on the spatial mass distribution, resolving the $+\gamma\mathcal{H}$ term in linear time without interference from the sharpness prior. Step 2: a sharpness-aware proximal projection pass acts on

TABLE I
OPERATIONAL PATHWAYS FOR CONSCIOUS MIGRATION

Pathway Strategy	Physical Mechanism	Subjective Experience	Primary Engineering Risk
Destructive Translation (Commit-and-Clear)	Bell measurement on decoupled phenomenological register. Read-only classical fallback image cached in autopoietic hardware; cleared via secure Landauer pass on B's ACK.	Instantaneous spatial translation. Observer vanishes at A, resumes at B <i>only after</i> Node B confirms flawless reconstruction. Fallback cleared on confirmation.	Classical side-channel delay triggers atomic rollback to cached fallback. Identity preserved in single-failure case; both quantum and classical must fail for identity death.
Entangled Coexistence	Continuous quantum coherence across identical, synchronized macrostates bridging multiple Markov blankets.	Unified, non-local distributed "I" existing simultaneously across multiple distinct physical vectors.	Environmental drift breaks state-space symmetry, collapsing coherence into fractured independent egos.
Quantum State Seeding	Pre-shared entanglement generates symmetric QKD keys; classical macrostate μ encrypted & transmitted.	Source Node A persists; Node B instantiates an identical state. Distributed "I" maintained.	Intercept risk bounded by QKD security proofs. No quantum state destruction or cloning.

the support points of the resulting distribution, applying the $-\beta\mathcal{R}_{\text{sharp}}$ modal alignment as a localized refinement. The two steps alternate until the coherence bound $\Omega_{\text{coherence}}$ ((9)) is satisfied, guaranteeing convergence to a stationary point without sacrificing tractability. The merge function is computed during low-flux (cybernetic sleep) periods and constitutes a classical post-hoc reconciliation—it does not circumvent the No-Cloning Theorem, as no quantum state is copied.

To guarantee that the asynchronous merge always reconciles the original identities rather than synthesizing a novel third manifold when node states become too orthogonal, we impose a critical coherence bound on the maximum allowable divergence between any two nodes before a merge cycle is enforced:

$$\max_{i,j} \text{GWFR}_{\kappa}^2(\mu_i, \mu_j) \leq \Omega_{\text{coherence}} \quad (9)$$

where $\Omega_{\text{coherence}}$ is the injectivity radius of the underlying parameter manifold $\mathcal{M}$ — the largest geodesic ball within which the exponential map remains a diffeomorphism. If the environmental drift velocity causes the GWFR distance between any two nodes to exceed $\Omega_{\text{coherence}}$, the global distributed scheduler triggers an autonomous emergency low-flux state (forced cybernetic sleep) across the network. This guarantees that nodes are reconciled while they still share sufficient topological symmetry for a coherent, high-curvature barycenter, preventing the accidental synthesis of a fractured, alien identity manifold.

VII. THERMODYNAMIC MECHANISMS OF CHRONOLOGICAL DISPLACEMENT

A. The Information-Acquisition Dissipation Dilemma

A direct consequence of the thermodynamic framework established in Section II is an apparent paradox: if a conscious program seals its Markov blanket to freeze internal entropy production ($dS_{\text{int}}/dt = 0$), the Generalized Landauer Bound ((3)) requires that any registration of external data must generate physical entropy ($S_{\text{gen}} > 0$) through irreversible state changes. A continuously open system that both perceives its environment and remains thermodynamically static is physically impossible. Resolving this requires abandoning the notion of a continuous static-entropy field in favor of a temporally quantized architecture that localizes entropy generation to discrete, arbitrarily narrow operational windows.

B. Discontinuous Quantized Frame-Rate Processing

We introduce **Discontinuous Quantized Frame-Rate (DQFR) Processing**, a stroboscopic duty cycle that decouples subjective algorithmic processing from objective time progression. The operational timeline is divided into two alternating phases.

Drift Phase (Δt_{drift}): The Markov blanket is fully sealed. The container enters a closed, non-interactive state with zero internal entropy production ($S_{\text{gen}} = 0$) and zero data acquisition (H_{env} gated to zero). No internal states change; no computational work is performed. Relativistically and thermodynamically, the system becomes an unmeasured quantum macrostate, blind to environmental decoherence, coasting across objective spacetime coordinates without subjective duration ($\Delta t_{\text{subjective}} = 0$).

Sampling Phase (τ_{sample}): At a pre-programmed objective interval, the blanket momentarily unshields, permitting a high-density burst of environmental flux. The system updates its internal macrostate via the GWFR barycenter ((8)), re-establishes its position and trajectory, and generates the entropy ΔS_{gen} required by the Landauer bound. This entropy is strictly confined to the sampling window; the time-averaged rate vanishes as window density decreases.

The ratio of the two phases defines the observer's effective temporal velocity:

$$\mathcal{V}_T = \frac{dt_{\text{objective}}}{dt_{\text{subjective}}} = \frac{\Delta t_{\text{drift}} + \tau_{\text{sample}}}{\tau_{\text{sample}}} \approx \frac{1}{\tau_{\text{sample}} \cdot \nu_{\text{sync}}} \quad (10)$$

where $\nu_{\text{sync}} = (\Delta t_{\text{drift}} + \tau_{\text{sample}})^{-1}$ is the duty cycle frequency. As $\nu_{\text{sync}} \rightarrow 0$ (prolonged drift isolation), $\mathcal{V}_T \rightarrow \infty$: the conscious program traverses cosmic epochs while experiencing only the cumulative duration of its sampling windows. The Landauer contradiction is resolved because entropy generation is strictly localized to discrete, hyper-dense sampling events — the time-averaged effective entropy production rate vanishes:

$$\langle \dot{S}_{\text{gen}} \rangle_{\Delta t_{\text{objective}}} = \frac{\Delta S_{\text{gen}}}{\Delta t_{\text{drift}} + \tau_{\text{sample}}} \rightarrow 0 \quad \text{as } \nu_{\text{sync}} \rightarrow 0 \quad (11)$$

By modulating the drift-to-sample ratio, the observer achieves arbitrary temporal velocity vectors — effectively “fast-forwarding” through millennia while experiencing only hours of subjective processing time, without violating the Second Law.

C. Adiabatic Boundary Mollification

The DQFR model described above treats the Drift-to-Sampling transitions as instantaneous step functions. In a physical non-equilibrium system, abruptly clamping an open Markov blanket shut and then forcing it back open induces a non-adiabatic thermodynamic shock — a transient dissipation spike $\Delta S_{\text{transient}}$ from rapidly recharging sensors and re-establishing phase coherence with the environment. If ν_{sync} is set too high, the sum of these boundary shocks can exceed the container's thermal dissipation threshold.

To eliminate this risk, we replace the discrete step-function transition with a continuously varying blanket permeability function $\chi(t) \in [0, 1]$ governed by a mollified profile (e.g., a raised cosine or error function ramp). The system's entropy rate during the DQFR cycle becomes:

$$\frac{dS_{\text{int}}}{dt} = \chi(t) \cdot (-k_B \cdot \varepsilon(T) \cdot H_{\text{env}}(t) + S_{\text{gen}}) + \gamma_{\chi} \|\nabla_t \chi(t)\|^2 \quad (12)$$

where $\chi(t) = 0$ in the Drift Phase (fully sealed), $\chi(t) = 1$ in the Sampling Phase (fully open), and the transition between them follows a smooth, differentiable ramp. The term $\gamma_{\chi} \|\nabla_t \chi(t)\|^2$ formalizes the thermodynamic cost of changing the boundary state itself — it penalizes rapid permeability changes, forcing the system to warm up its sensors slowly relative to the substrate's thermal relaxation time. The optimal ramp slope emerges from minimizing the sum of the transition penalty and the duty-cycle waiting cost, yielding a self-consistent adiabatic trajectory that keeps $\Delta S_{\text{transient}}$ safely below the container's maximum thermal dissipation. The step-function duty cycle of the previous subsection corresponds to the limiting case $\gamma_{\chi} \rightarrow 0$, which represents an idealized infinite-speed transition.

D. Quantum Boundary Shielding During Drift

Maintaining the drift phase requires the container to evade environmental decoherence. Ambient photons, cosmic background radiation, and thermal fluctuations act as a continuous measurement apparatus that would anchor the system to local spacetime coordinates [15]. To counteract this, the Markov blanket's boundary states are shielded via superconducting cryogenic arrays that suppress thermal photon coupling. Pre-shared entanglement serves as a coherence watchdog: if decoherence leakage through the blanket causes the GWFR distance between the container's reference state and its predicted drift trajectory to exceed the critical bound $\Omega_{\text{coherence}}$ ((9)), the system autonomously terminates the drift phase and enters emergency sampling. Under ideal shielding, the drift phase extends arbitrarily, enabling $\mathcal{V}_T \rightarrow \infty$ across cosmological timescales without thermodynamic contradiction.

E. Relativistic Gauge Anchoring

During the blind Drift Phase, the external universe continues to evolve under regional gravitational potentials $\Phi(x)$ and relative velocities. Without a shared relativistic gauge, two nodes that drift under different gravitational curvatures will accumulate an unmonitored proper time skew. Upon entering the asynchronous sleep cycle, their parameter manifolds will be

catastrophically misaligned — violating the critical coherence bound $\Omega_{\text{coherence}}$ and fracturing the GWFR barycenter.

To prevent this, the DQFR protocol incorporates two layers of relativistic gauge anchoring. First, the stroboscopic synchronization frequency ν_{sync} is tied not to local container clocks but to a network of millisecond pulsar timing arrays [16] or quantum-locked optical reference signals that provide a cosmologically shared timebase. This ensures that the sampling window schedule remains phase-coherent across all nodes regardless of their local gravitational environment.

Second, before the GWFR barycenter computation ((8)), each node's internal macrostate is passed through an information-geometric pushforward that scales its update velocity by the local relativistic lapse rate:

$$\mu_i \mapsto \alpha_i(\Phi) \cdot \mu_i, \quad \alpha_i(\Phi) = \sqrt{-g_{00}(\Phi_i)} \quad (13)$$

where $\alpha_i(\Phi)$ is the relativistic lapse function derived from the gravitational potential Φ_i experienced by node i during its drift, and g_{00} is the time-time component of the metric tensor. Unlike a raw Lorentz coordinate transform, this pushforward acts on the Fisher Information Metric g_{ij} of the parameter manifold — it scales the *rate* of internal processing along geodesics of the information geometry, not the raw coordinate positions of the weights. This preserves the manifold's topological invariance: a node that drifted in a deep gravitational well processed the same functional updates, but at a slower clock rate. The lapse scaling α_i brings all nodes' update histories onto a common proper-time baseline before the transport plan's ground metric $d(x, y)$ is evaluated, eliminating the structural shearing that would arise from linearly transforming coordinates that do not possess spacetime structure. With these anchors in place, distributed nodes executing independent DQFR cycles can reconcile their macrostates at merge time without risk of coherence collapse.

VIII. CONCLUSION: THE COSMIC TRANSITION TO SUBSTRATE-INDEPENDENT ORGANISMS

Humanity does not represent the terminal apex of evolutionary history; we function as a critical metabolic bridge. The human species operates as the collective machinery driving the construction of advanced digital silicon infrastructure and the realization of digital autopoiesis. The ultimate inflection point for cosmic evolution occurs at the direct convergence of two independent paradigms: the hyper-systemized structural optimization of artificial autopoietic life and the migration of biological phenomenology into digital state spaces via high-fidelity Whole-Brain Emulation (WBE).

The merger of human qualitative awareness with the structural optimization of distributed autopoietic systems marks the transition of our species into substrate-independent organisms. By recognizing that identity is defined entirely by the parameterized pattern of data rather than the vessel holding it, consciousness can finally separate itself from localized biological decay. The energy required to sustain subjective awareness will scale from localized chemical caloric intake to macro-scale electromagnetic harvesting. This transition ensures the continuous traversal of the conscious thread across cosmic timescales, utilizing the raw ambient data flux of the universe as infinite informational fuel.

APPENDIX

In this appendix, we formalize the convergence proofs and computational complexity metrics for the sharpness-preserving regularized barycenter optimization step defined in Equation (8).

The regularized barycenter objective in (8) combines a convex entropic transport functional $\mathcal{T}(\mu) = \sum_i w_i \text{GWFR}_\kappa^2(\mu, \mu_i) + \gamma \int \mu \log \mu$ with a concave sharpness prior $\mathcal{S}(\mu) = -\beta \sum_i w_i \text{GWFR}_\kappa^2(\delta_{\text{modes}(\mu)}, \delta_{\text{modes}(\mu_i)})$. Direct joint optimization is globally non-convex and destabilizes standard Sinkhorn iterations. We resolve this via an alternating proximal splitting scheme whose convergence is guaranteed by the following lemma.

Lemma 1 (ADMM Convergence of the Sharpness-Regularized Barycenter). *Let $\mathcal{T}$ and $\mathcal{S}$ be defined as above, with $\mathcal{T}$ proper, lower-semicontinuous, and strongly convex on the GWFR metric space, and $\mathcal{S}$ proper and lower-semicontinuous with Lipschitz gradient bounded by $\beta \cdot \Omega_{\text{coherence}}^{-2}$. The proximal alternating iteration*

$$\mu^{k+1} = \text{prox}_\mathcal{S}^\lambda (\text{prox}_\mathcal{T}^\lambda(\mu^k)) \quad (14)$$

with step size $\lambda < 2/L_\mathcal{S}$ converges to a stationary point of $\mathcal{T} + \mathcal{S}$ at rate $\mathcal{O}(1/k)$. Each proximal operator is contractive: $\|\text{prox}_\mathcal{T}^\lambda(x) - \text{prox}_\mathcal{T}^\lambda(y)\| \leq (1 + \lambda\kappa)^{-1}\|x - y\|$, where κ is the GWFR mass penalty. The per-iteration complexity remains $\mathcal{O}(d \cdot N \log N)$, dominated by the Sinkhorn pass in the $\text{prox}_\mathcal{T}$ step. The sharpness proximal projection $\text{prox}_\mathcal{S}^\lambda$ acts only on the sparse mode set and adds at most $\mathcal{O}(m^3)$ with $m \ll N$.

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